WAYAMBA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (GENERAL/JOINT MAJOR/ SPECIAL) Degree Examination
Level 3 Semester I– (Academic Year 2022/2023) Feb/Mar 2025

Department of Computing and Information Systems

CMIS 3142 – Computational Methods

No. of Pages: Three (03) *No. of Questions: Six (06)* *Time: Two (02) hours*

Answer four Questions only

Calculators are permitted

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1)

a) Describe overflow and underflow when representing a floating point number using IEEE 754 format.

b) What is the special feature in the Denormalized number in the IEEE 754 format?

c) Convert the following numbers into IEEE 754 single precision format

i) 256 ii) 1023 iii) 364

d) Write a Java program to convert an octal number to hexa-decimal number and vice versa.

2) Suppose a student has completed one course unit with 2 – credits, three course units each of 3 – credits and two course units each of 1-credit with grades, A, C, B, D, B- and A+ respectively. The calculation of GPA is given below:

$$\text{GPA} = \frac{(2*4.0)+(3*2.0)+(3*3.0)+(3*1.0)+(1*2.7)+(1*4.0)}{2+3+3+3+1+1}$$

$$= \frac{32.7}{13} = 2.5153846$$

$$= 2.52$$

GPA is considered to be rounded up to two decimal points.

Assume that a student should complete forty course units each of 3 credits to complete a Degree programme.

a) Describe, with suitable examples, the impacts of rounding off error in the GPA calculation.

b) Describe, with suitable examples, the impacts of chopping error in the GPA calculation.

c) Write a Java programme to perform the following tasks:

   i. Read the module codes and credit values.

   ii. Calculate the GPA value.

   iii. Calculate the round off error and chopping error while calculating the GPA.

   iv. Display the difference between the round off error and chopping error.

3)

a) Describe an efficient way to represent a sparse symmetric matrix to minimize the memory use in a computer program.

b) Write a Java method to read and store elements of a sparse symmetric matrix.

c) Write a Java method to return the element at a given row and column of a sparse symmetric matrix.

d) Write a Java method to display only the stored elements of the sparse symmetric matrix and the row number and the column number of each element.

e) Write a Java method to display all the elements (including zeros) of a sparse symmetric matrix in the form of a matrix.

4)

a) What is the main difference between Jacobi method and Gauss-Seidel method?

b) There are three students called Akila, Bala and Pathma and they have 15, 28 and 18 rupees respectively.

   Akila bought 8 pine apples, 2 mangos and 1 apple;
   Bala bought 4 pine apples 9 mangos and 2 apples;
   Pathma bought 1 pine apple, 1 mangos and 5 apples;

   i) Formulate a system of linear equations to find the price of each fruit.

   ii) Solve the above problem using Jacobi method for which you may use the initial values (1, 1, 2) respectively. You may go for a maximum of 10 iterations.

   iii) Solve the above problem using Gauss Seidel method for which you may use the initial values (1, 1, 2) respectively. You may go for a maximum of 6 iterations.

   iv) Comment on the performance of both of those methods.

c) Suppose you have a Java program to solve a system of linear equations using Jacobi method. Describe the way to modify the computer program for solving the system of linear equation using Gauss Seidel method.

5)

a) Solve the following equations using SOR (Successive over-relaxation) method (with initial guess (0,0,0) and w = 1.25)

i) $15x + 2y + z = 22$
ii) $4x + 15y + 2z = 40$
iii) $x + y + 15z = 48$

b) Write a Java program to solve a system of linear equations using Cholesky method. Describe the essential conditions to be satisfied to apply the Cholesky method.

c) Solve the following system of equations using the Cholesky decomposition method

i) $8x + 2y + 3z = 13$
ii) $2x + 10y + 4Z = 16$
iii) $3x + 4y + 9z = 16$

6)

a) Describe the Gaussian elimination method to solve a given system of liner equations.

b) Write program in any programming language to solve a system of linear equations using Gaussian elimination.

c) Find the solution to $f(x) = 4x^3 - 3x^2 - 5x - 10$ using the fixed iteration method.